

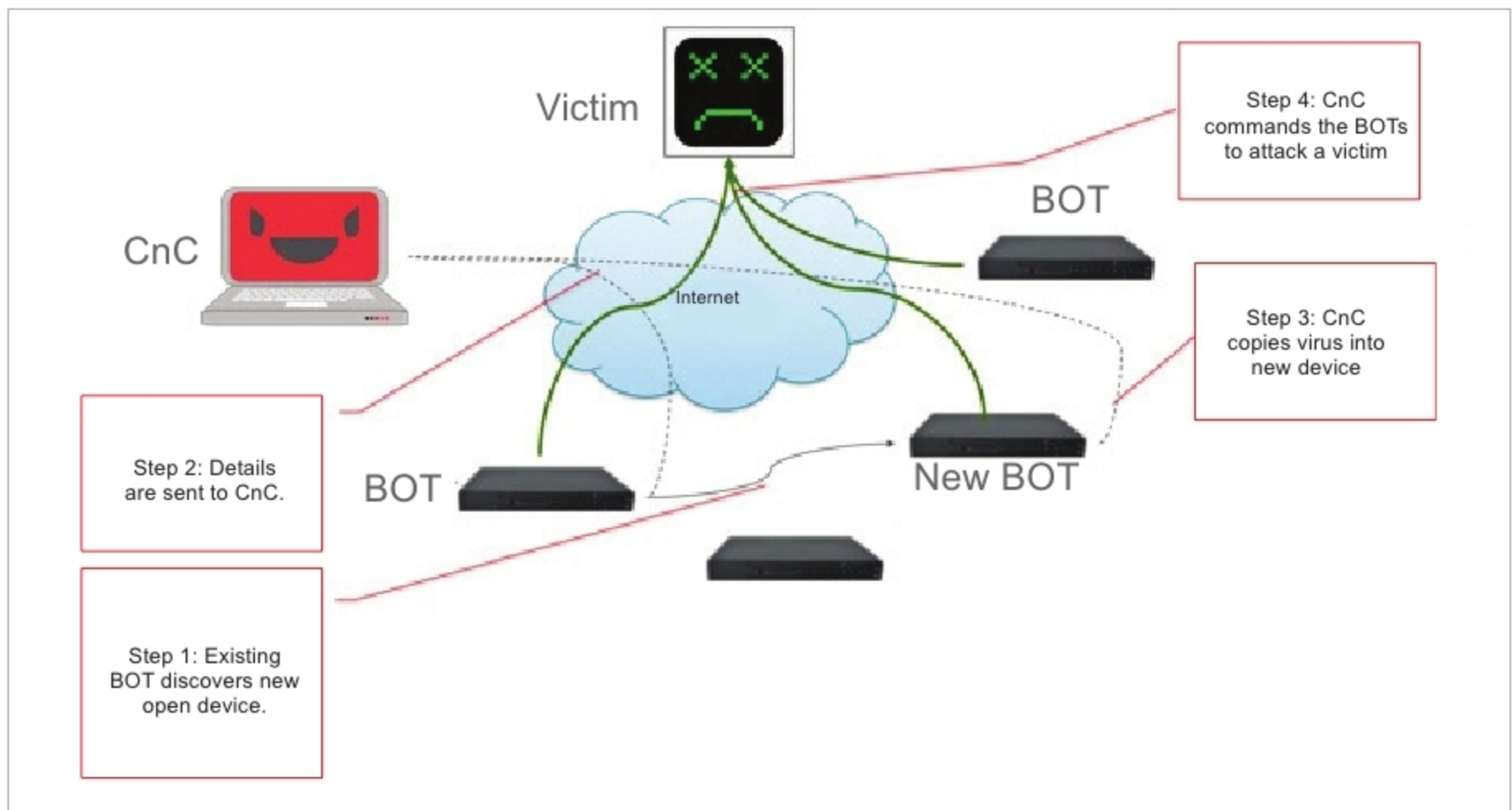


Figure 1: The Mirai botnet

purpose of this bot is to brick devices so they will not be usable once attacked. (<https://www.bleepingcomputer.com/news/security/brickerbot-author-retires-claiming-to-have-bricked-over-10-million-iot-devices/>)

Note: Bricking is the process of changing the code of the device so that the hardware can no longer be used, thereby turning the device essentially into a brick (totally unusable).

The Mirai botnet attack: The Mirai botnet attack took place in 2016. This was a major malware/virus attack. Mirai is a self-propagating botnet virus. The source code for it was made publicly available by the author after a successful and well-publicised attack on the Krebs website. Since then, the source code has been built and used by many others to launch attacks on Internet infrastructure, causing major damage. The Mirai botnet code infects poorly protected Internet devices by using telnet to find those that are still using their factory default usernames and passwords. The effectiveness of Mirai is due to its



Figure 2: Princeton IoT Icon

ability to infect tens of thousands of these insecure devices and co-ordinate them to mount a DDoS attack against a chosen victim. (<https://www.corero.com/resources/ddos-attack-types/mirai-botnet-ddos-attack>)

Detection and prevention of security attacks

The major cause of concern in IoT devices is the use of the Linux OS by the device vendors instead of developing a custom OS, which is time consuming and expensive. Most of the attackers/hackers know this and so target these devices.

Some time back, Symantec came up with a solution for this by using a router called Norton Core (<https://us.norton.com/core>). This was not a success as it was expensive and had a monthly maintenance cost. In addition, people felt that it was too early to use such a router that came with a monthly subscription, since most homes still do not have enough IoT enabled devices to make such an investment worthwhile.

Open source security tools

Subsequently, many state-of-art security tools with multiple security features have been launched. Some of the most used and popular open source security tools are featured below.

Princeton IoT-Inspector

This is an open source desktop tool with a one-click, easy-to-install process. It has many built-in security validation features:

- Automatically discovers IoT devices and analyses their network traffic.
- Helps one to identify security and privacy issues with graphs and tables.
- Requires minimal technical skills and no special hardware.